ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION â€" DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING â€" SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted. Mobile phones are strictly prohibited.

--- UNIT I ---

Q.1 (a) What is 3-Tier Database Architecture? Explain Data Independence (Logical and Physical). [7 Marks]
(b) Design an Entity-Relationship (ER) diagram for an Online Hospital Management System. Show strong/weak entities, primary keys, and cardinality ratios. [7 Marks]

[OR]

Q.1 (Alt) Alternative: (a) Convert the above ER diagram into Relational Tables with appropriate integrity constraints. [7 Marks]
(b) Explain Generalization, Specialization, and Aggregation in Enhanced ER (EER) modeling. [7 Marks]

--- UNIT II ---

Q.2 (a) Explain fundamental Relational Algebra operations: Select, Project, Cartesian Product, Union, Set Difference, and Rename with examples. [7 Marks]
(b) Given schemas Employee(eid, ename, dept, salary) and Department(did, dname, budget), write SQL queries for aggregate functions, nested subqueries, and group-by having clauses. [7 Marks]

[OR]

Q.2 (Alt) Alternative: (a) Differentiate between Tuple Relational Calculus (TRC) and Domain Relational Calculus (DRC). [7 Marks]
(b) Explain Views, Triggers, and Stored Procedures in PL/SQL with syntax and use cases. [7 Marks]

--- UNIT III ---

Q.3 (a) What is Normalization? Explain Functional Dependency and Armstrong's Axioms. [7 Marks]
(b) Explain 1NF, 2NF, 3NF, and BCNF with concrete counterexamples and decomposition steps. [7 Marks]

[OR]

Q.3 (Alt) Alternative: (a) Given relation R(A, B, C, D, E) and FDs {A->BC, CD->E, B->D, E->A}, find candidate keys and highest normal form. [7 Marks]
(b) Explain Lossless-Join Decomposition and Dependency Preservation with proofs. [7 Marks]

--- UNIT IV ---

Q.4 (a) State ACID properties of database transactions. Explain each property with concurrency scenarios. [7 Marks]
(b) Explain Conflict Serializability and View Serializability. How does Precedence Graph test conflict serializability? [7 Marks]

[OR]

Q.4 (Alt) Alternative: (a) Describe Concurrency Control protocols: Two-Phase Locking (2PL), Strict 2PL, and Rigorous 2PL. [7 Marks]
(b) Explain Timestamp-based ordering and Thomas' Write Rule. [7 Marks]

--- UNIT V ---

Q.5 (a) Explain Database Recovery Techniques: Log-Based Recovery (Deferred and Immediate modification), Checkpoints, and Shadow Paging. [7 Marks]
(b) Differentiate between B-Tree and B+ Tree indexing. Why are B+ Trees preferred for database indices? [7 Marks]

[OR]

Q.5 (Alt) Alternative: (a) Explain Query Processing steps: Parsing, Translation, Optimization, and Evaluation. [7 Marks]
(b) Write short notes on: (i) ACID vs BASE in NoSQL (ii) Deadlock prevention in databases. [7 Marks]

**** END OF QUESTION PAPER â€" MITS GWALIOR AUTONOMOUS EXAMINATION SYSTEM ****